

Validation of temporal synchronization between two Maxi-Phyling using GPS time

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Abstract: Synchronizing sensors within the same system poses a recurring challenge and is essential to obtain accurate measurements. The aim of this study is to validate the temporal synchronization between two Maxi-Phyling GPS units (Phyling, France) using GPS time. We compare two versions of our GPS units (V2.2 and V2.3). The GPS time difference between two Maxi-Phyling V2.3 averages 6.05 ms ($\pm 6,64$ ms), with a maximum difference of 24.7 ms. In contrast, the difference between a V2.1 and a V2.2 is 500 ms (± 230 ms) on average, with a maximum of 800 ms. In conclusion, it is feasible to achieve temporal synchronization with high accuracy for two Maxi-Phyling (V2.3) using the GPS time data.

1. Introduction

Sensor systems are characterized by the simultaneous utilization of multiple sensors. In sports science, the combined use of sensors is common to enhance performance analysis. For instance, an Inertial Measurement Unit (IMU) system can yield more intricate data compared to a single IMU sensor. IMU systems enable the determination of angular joints or segmental positions [1].

For system measurements to be valid, time synchronization between sensors is a critical factor. In wired networks, it is easier to achieve accurate time synchronization compared to wireless systems. Several techniques are used in networks.

Wired synchronization

The time synchronization of a wired sensor network is an accurate solution with minimal cost. However, this type of synchronization imposes constraints on sensors placements and manipulability. For instance, the Transistor-Transistor-Logic (TTL) port is utilized, employing simultaneous digital pulses for synchronization purposes [2].

Wireless synchronization

Time synchronization of a wireless sensor network is a more complex task. Several techniques exist with different accuracy. GPS time appears to be a reliable method for synchronizing wireless sensors equipped with GPS units, particularly when the sky is unobstructed [3].

The aim of this study is to validate the accuracy of time synchronization between two Maxi-Phyling sensors using GPS time.

2. Methods

The Maxi-Phyling is an acquisition unit with integrated sensors such as GPS (10 Hz) and IMU (200 Hz). In this study, we will compare GPS time accuracy between three Maxi-Phyling with different hardware version (2 Maxi-Phyling V2.3 and 1 older generation Maxi-Phyling V2.2). The V2.3 version is equipped with M10Q GPS chip while V2.2 has M8N. Specifically, units numbered 189 and 190 are the newest version (V2.3) while unit number 106 is the Maxi-Phyling unit with the older version (V2.2).



Figure 1 : Maxi-Phyling unit

To precisely compare GPS time differences between units, we equipped all Maxi-Phyling units with a TTL port (1kHz). A pushbutton was connected to each unit and utilized to transmit digital pulses simultaneously.

Twelve digital pulses were transmitted simultaneously to the three units at a frequency of 1kHz over a total duration of 10 minutes. The GPS time at each pulse is compared between Maxi-Phyling units.

3. Results

The mean and maximum GPS time differences between Maxi-Phyling units are presented in Table 1.

Table 1 : Mean and maximums GPS time difference between Maxi-Phyling units.

	Mean ($\pm$ SD)	Maximum
Δ189-190	6,08 $\pm$ 6,64	24,70
Δ190-106	505,22 $\pm$ 233,27	863,28
Δ189-106	507,98 $\pm$ 230,23	863,18

4. Discussion

The aim of this study was to assess the accuracy of GPS time synchronization between Maxi-Phyling units. According

to the results presented in Table 1 the average difference between two Maxi-Phyling V2.3 is 6,08 ms. In the analysis of 12 TTL pulses, the maximum GPS time difference between these two latest versions was 24,70 ms.

In contrast, when using a Maxi-Phyling V2.2 and a Maxi-Phyling V2.3, the GPS time difference was on average 505 ms or 507 ms with a maximum difference of 863 ms. These results highlight the improved accuracy of the new Maxi-Phyling version for GPS time accuracy. This good accuracy could be utilized for temporal synchronization purposes, for example when using several Maxi-Phyling sensors and other sensors equipped with a GPS chip (ie. camera, etc).

In futures studies, we will conduct a comparative analysis of Maxi-Phyling versions to assess their accuracy in GPS positioning.

5. Conclusion

Thanks to the new version of Maxi-Phyling (V2.3), it is possible to synchronize our GPS units with high accuracy.

6. Bibliography

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